

E-Waste & Environmental Management

Activity Worksheet

The Hidden Carbon Footprint of Our Gadgets

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Date: 05/08/2026

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Part A: Carbon Footprint Calculation

Use the carbon footprint calculator provided by your faculty.

Parameter	Your Response
Annual Carbon Footprint	150 tCO ₂ e / year
National Average	50 tCO ₂ e / year
Global Average	140 tCO ₂ e / year
Number of Earths Required (if provided)	1.07 Earths

Which category contributed the most?

☐ Transportation ☐ Home Energy ☒ Food ☐ Shopping & Consumption

☐ Electronics / Digital Lifestyle ☐ Other: _____

Which result surprised you the most?

Food contribution to the Carbon footprints surprised me the most.

Part B: My Electronic Lifestyle

1. How many smartphones have you owned in the last 10 years?

☐ 1 ☒ 2-3 ☐ 4-5 ☐ More than 5

2. How often do you replace your smartphone?

☐ Every year ☐ Every 2 years ☐ Every 3-4 years ☒ More than 5 years

3. What do you usually do with your old phone?

☒ Keep ☐ Sell ☐ Exchange ☐ Donate

☐ Recycle ☒ Throw away

4. Have you ever repaired an electronic device instead of replacing it?

☒ Yes ☐ No

If yes: _____

5. Have you ever recycled e-waste through an authorized recycler?

☐ Yes ☒ No

Item: _____

Unused Electronic Items at Home	Number
Mobile phones	<u>0</u>
Chargers	<u>0</u>
Earphones	<u>2</u>
Batteries	<u>0</u>
Cables	<u>3</u>
Other	<u>0</u>

Part C: Think Like an Environmental Engineer

Habit	Higher Carbon Footprint	More E-Waste
Buying a new phone every 2 years	☒	☒
Repairing a laptop	☐	☐
Keeping old phones unused in drawers	☒	☐
Recycling electronic devices	☐	☐
Using devices for 5–6 years	☐	☐

Which stage of an electronic product's life cycle contributes the MOST carbon emissions?

☐ Raw material extraction ☒ Manufacturing ☐ Transportation

☒ Usage ☒ Disposal

Why is extending the life of an electronic device environmentally beneficial?

→ It helps to reduce e-waste and carbon foot prints.

It makes us more aware of towards environment pollution.
save expense.

Part D: Reflection

Three habits that increase my personal carbon footprint:

- ① using personal transport
- ② Not recycling food & other substances
- ③ Using too much electrical appliances.

Three habits that increase e-waste generation:

- ① Keeping spare e-gadgets
- ② Getting attracted to new gadgets & buying them
- ③ Using

Action to Reduce Carbon Emissions & E-waste	Expected Impact
Use public transport or carpool	Reduces carbon emission & fuel consumption
Repair and use electronic devices for a longer time	Reduces e-waste and saves natural resource
Recycle old electronics through authorized collection centre	Prevents pollution and recovers valuable materials

Discussion Questions (Optional)

1. Which stage of an electronic product's life cycle has the highest carbon footprint and why?

Manufacturing stage has the highest carbon footprint because extracting raw materials, producing components and assembly consumes large amount of energy and resource.

2. How does extending the lifespan of smartphones reduce both carbon emissions and e-waste?

Using smartphones for a long time reduces the need to manufacture new devices, which lowers carbon emission and decrease the amount of electronic waste.

3. Why is recycling alone not enough for sustainable e-waste management?

Recycling cannot recover all materials and also uses energy. Reducing, reusing and repairing electronic devices are equally important to minimize waste.

4. Explain how the 5Rs contribute to lowering the environmental impact of electronic products.

→ The 5Rs - Refuse, Reduce, Reuse, Repair and

Recycle - helps conserve resources, reduce pollution, extend product life and minimize e-waste.

5. As an engineering student, what innovative solution would you propose to reduce the carbon footprint associated with electronic devices?

I would develop an AI based smart e-waste management system that helps user, repairs, reuse & donate & recycle electronic devices efficiently & reducing unnecessary purchase and lowering carbon emission.

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